# An Internship Report

Of

# Tata iQ — Data Analytics Virtual Internship

Submitted

To

**School of Computer Science and Engineering**

**IILM University**

In partial fulfilment of the requirement of degree of

**B.Tech (Computer Science and Engineering)**

By

**Name of Student: PRASHANT SHARMA**

**Roll Number of Student: 25SCS1003004867**

**Section: 2CSE37**

**Batch: 2025–2029**

# CANDIDATE'S DECLARATION

I, PRASHANT SHARMA, do hereby declare that the internship report titled “Tata iQ — Data Analytics Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

**Student Name: PRASHANT SHARMA**

**Roll No.: 25SCS1003004867**

**Section: 2CSE37**

# ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank The Forage and Tata Consultancy Services (Tata iQ) for providing me with this exceptional opportunity through the Data Analytics Virtual Experience Program. The structured, task-based curriculum gave me hands-on exposure to real-world data analytics workflows — from exploratory data analysis and predictive modelling to stakeholder communication and responsible AI design. I am also thankful to my institute and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

**Student Name: PRASHANT SHARMA**

**Roll No.: 25SCS1003004867**

**Section: 2CSE37**

# TABLE OF CONTENTS

# 3. INTERNSHIP COMPLETION CERTIFICATE

TATA

Forage

Inspiring and empowering future professionals

## Prashant Sharma

## GenAI Powered Data Analytics Job Simulation

Certificate of Completion

September 9th, 2026

Over the period of September 2026, Prashant Sharma has completed practical tasks in:

Exploratory data analysis and risk profiling
Predicting delinquency with AI
Business report and data storytelling for collections strategy
Implementing an AI-driven collections strategy

**Tom Brunskill**
Co-Founder of Forage

Enrolment Verification Code 6aa0315939a1f8e66607cab1 | User Verification Code 6aa02cd139a1f8e66606d5d5 | Issued by Forage

# 4. PROJECT DESCRIPTION

## 4.1 Introduction

This report presents a detailed account of the Data Analytics Virtual Internship undertaken through The Forage platform, in collaboration with Tata Consultancy Services (Tata iQ). The internship was carried out from August 2026 to September 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme.

The internship was structured around a realistic business scenario centred on Geldium, a fictional fintech company facing a significant collections challenge. Working as a data analyst embedded with Tata iQ's analytics team, I was tasked with analysing Geldium's customer dataset, building a framework for a delinquency risk prediction model, developing actionable stakeholder recommendations, and designing a responsible, AI-powered collections system. The work spanned the full analytics value chain — from raw data exploration through to executive-level strategic presentation.

The internship followed a structured, task-based curriculum. On successful completion of all tasks, a Virtual Internship Certificate was awarded by The Forage in association with Tata Consultancy Services.

## 4.2 Organization Profile

Tata Consultancy Services (TCS) is a global leader in IT services, consulting, and business solutions, headquartered in Mumbai, India. With a presence in over 50 countries and a workforce of more than 600,000 professionals, TCS is one of the world's largest IT companies and a flagship company of the Tata Group. TCS partners with The Forage to offer the Tata iQ Data Analytics Virtual Experience Program — an industry-aligned, task-based program designed to give students hands-on exposure to the kind of data analytics work performed within TCS's advanced analytics practice (Tata iQ). The program covers the full analytics workflow: from exploratory data analysis and predictive model design, through stakeholder reporting, to responsible AI strategy — mirroring real client engagements in financial services and credit risk.

### 4.3 Problem Statement

Geldium, a fintech lending company, was experiencing a growing collections problem: a significant and rising proportion of its loan customers were becoming delinquent — failing to make required payments on time. The collections team was operating reactively, reaching out to customers only after a payment had already been missed, which was proving both costly and ineffective.

The challenge presented to the Tata iQ analytics team was threefold: (1) to understand the customer data landscape through rigorous exploratory analysis and identify the key drivers of delinquency risk; (2) to design a predictive model framework capable of identifying at-risk customers before a payment is missed, enabling proactive intervention; and (3) to translate those insights into a strategic recommendation for an autonomous, responsible AI-powered collections system that is fair, explainable, and compliant with regulatory requirements.

### 4.4 Project Objectives

- To conduct a thorough Exploratory Data Analysis (EDA) of Geldium's 500-customer dataset, assessing data quality, identifying missing values, standardising inconsistent fields, and surfacing early risk indicators.
- To analyse correlations between financial, demographic, and behavioural variables and identify the top predictors of delinquency risk.
- To design and justify a predictive model framework — recommending gradient-boosted decision trees (XGBoost/LightGBM) with logistic regression as a transparent baseline — and define an evaluation strategy using AUC-ROC, F1 Score, Precision, Recall, and a Confusion Matrix.
- To develop SMART, evidence-based recommendations for Geldium's Head of Collections, translating predictive insights into targeted, actionable intervention strategies for identified at-risk customer segments.
- To design a high-level AI-powered collections system that describes the system workflow (inputs, decision logic, actions, learning loop), the role of agentic AI, responsible AI guardrails (fairness, explainability, compliance), and expected business impact.
- To present findings and the AI system design as a polished executive-level PowerPoint deck, simulating a real business briefing.

### 4.5 Scope of the Project

The scope of this internship spans the complete data analytics workflow for a credit risk / collections use case. It begins with dataset profiling and missing-data treatment on a 500-record customer dataset (containing financial, demographic, payment-behaviour, and account-tenure fields), progresses through feature correlation analysis and risk-indicator identification, then moves to predictive model design and evaluation-strategy definition. The scope further extends to stakeholder communication — translating technical findings into business-ready recommendations — and culminates in a strategic AI system design covering agentic automation, responsible-AI governance, and quantitative and qualitative business impact projections. The internship was conducted entirely within The Forage's virtual, task-based environment and did not involve deployment on any live organizational production system.

### 4.6 Technologies and Tools Used

- Data Analysis & EDA: Python (Pandas, NumPy), statistical correlation analysis, missing-data profiling
- Predictive Modelling: Gradient-Boosted Decision Trees (XGBoost / LightGBM), Logistic Regression, Scikit-learn
- Model Explainability: SHAP (Shapley Additive Explanations), Feature Importance Rankings
- Evaluation Metrics: AUC-ROC, F1 Score, Precision, Recall, Confusion Matrix
- Data Handling: Median imputation (global and group-wise), Employment-Status standardisation, Imbalanced-data awareness
- Visualisation & Reporting: Microsoft PowerPoint, Microsoft Word
- Learning Platform: The Forage — Tata iQ Data Analytics Virtual Experience

### 4.7 System Architecture

The proposed AI-powered collections system for Geldium follows a layered architecture that takes customer data as input, scores each customer for delinquency risk, routes them to an appropriate intervention track, learns from outcomes, and escalates edge cases for human review. The six core layers are:

- Data Ingestion Layer: Customer demographic, financial, account, and payment-history data is ingested from Geldium's core systems on a scheduled or event-driven basis.
- Feature Engineering Layer: Raw fields are transformed into model-ready features — credit utilization ratio, payment trend (direction and magnitude over 6 months), income-to-debt ratio, and account-tenure band.
- Scoring Layer: The trained gradient-boosted model scores each customer with a 0–1 probability of delinquency, bucketed into Low / Medium / High risk tiers using calibrated thresholds.
- Decision & Action Layer: Risk tier and top contributing SHAP factors are surfaced to the collections platform; the system autonomously routes Low-risk customers to a self-serve nudge, Medium-risk to a courtesy outreach, and High-risk to an agent-assisted payment-plan workflow.
- Human-in-the-Loop Layer: Complex or ambiguous cases, disputed decisions, and any customer who requests human review are escalated to a human collections specialist before any adverse action is taken.
- Learning Loop Layer: Outcomes (payment made / delinquency confirmed) are fed back to retrain the model on a regular cadence, closing the feedback loop and keeping predictions calibrated to changing customer behaviour.

## 4.8 Methodology

The internship followed a structured, task-based methodology combining analysis, modelling design, stakeholder communication, and strategic AI design:

| Task | Module | Description |
|---|---|---|
| Task 1 | Exploratory Data Analysis | Profiled 500-record customer dataset; identified and treated missing values across 6 variables; standardised Employment_Status; surfaced top risk indicators through correlation analysis. |
| Task 2 | Predictive Model Framework | Recommended gradient-boosted decision trees (XGBoost/LightGBM) with logistic regression baseline; defined top 5 input features; designed 5-step model workflow and evaluation strategy (AUC-ROC, F1, Precision, Recall, Confusion Matrix). |
| Task 3 | Business Summary & Stakeholder Report | Translated predictive insights into a SMART recommendation for Geldium's Head of Collections; identified three high-impact customer segments; proposed a proactive early-outreach program targeting 70%+ credit utilization trend. |
| Task 4 | AI-Powered Collections Strategy (Executive PPT) | Designed a full AI-powered collections system covering system workflow, agentic AI roles, responsible-AI guardrails, and expected business impact; delivered as a polished executive PowerPoint briefing. |

### 4.9 Expected Outcomes

- Practical ability to profile and clean a real-world tabular dataset — treating missing values, resolving data quality issues, and identifying risk-relevant features through EDA.
- Understanding of how to design and justify a predictive model for a financial services use case, including the trade-offs between model complexity, accuracy, and regulatory interpretability.
- Capability to translate technical model outputs into clear, SMART business recommendations for non-technical stakeholders (e.g., Head of Collections).
- Skill to communicate findings through a professional, executive-level PowerPoint presentation that balances data rigour with strategic narrative.
- Knowledge of responsible AI principles in a financial context — including fairness metrics (demographic parity, disparate impact), explainability tools (SHAP), and regulatory compliance requirements.
- Completed all four tasks and received the Tata iQ Data Analytics Virtual Internship Certificate from The Forage.

### 4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by The Forage in association with Tata Consultancy Services (Tata iQ), confirming successful completion of the Data Analytics Virtual Experience Program, is attached in Chapter 3 of this report. Copies of task submission confirmations and any skill badges earned on The Forage platform are attached below as further communication proof.

# 5. BIBLIOGRAPHY/REFERENCES


1. The Forage, “Tata iQ — Data Analytics Virtual Experience Program,” Program ID: gMTdCXwDdLYoXZ3wG, 2026. [Online]. Available: https://www.theforage.com/virtual-experience/gMTdCXwDdLYoXZ3wG/tata/data-analytics-t3zr/intro-scenario

2. Tata Consultancy Services, “Tata iQ Analytics Practice,” 2026. [Online]. Available: https://www.tcs.com

3. Chen, T. and Guestrin, C., “XGBoost: A Scalable Tree Boosting System,” in Proc. KDD, 2016.

4. Lundberg, S. M. and Lee, S.-I., “A Unified Approach to Interpreting Model Predictions (SHAP),” Advances in Neural Information Processing Systems, 2017.

5. Scikit-learn Documentation, https://scikit-learn.org

6. TensorFlow Documentation, https://www.tensorflow.org/

7. The Forage Help Centre, https://www.theforage.com